

EXAM 1 – March 11, 2025

ISE 5970: Advanced Systems Optimization

University of Oklahoma

College of Engineering

School of Industrial and Systems Engineering

Professor: Andrés D. González

The exam is open-note and open-book, but you cannot use external help from any person or AI (e.g., ChatGPT).

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Pledge: “On my honor, I have neither given nor received inappropriate assistance in the completion of this Exam.”

Student signature: Pepi Perez

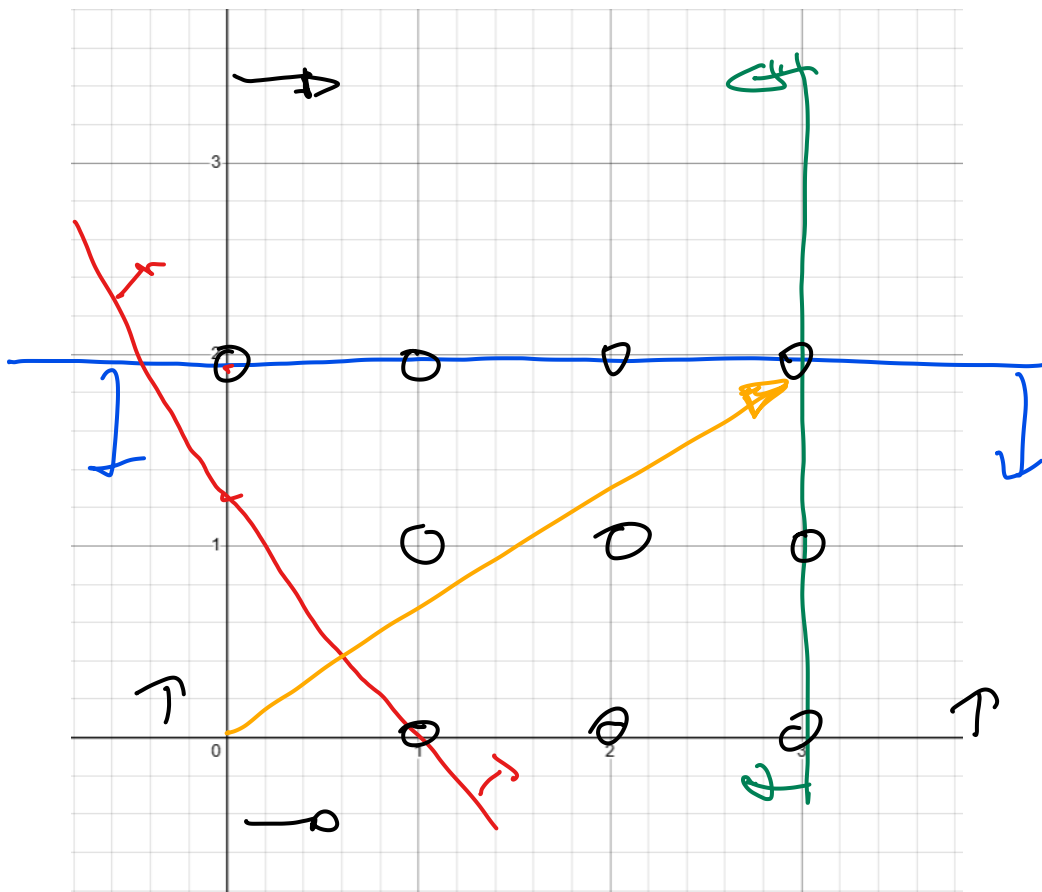
Assume you have the following integer problem (IP):

$$\begin{aligned} \min & 3x_1 + 2x_2 \\ \text{s. t. } & 5x_1 + 4x_2 \geq 5 \\ & x_1 \leq 3 \\ & x_2 \leq 2 \\ & x_1, x_2 \in \mathbb{Z}^+ \cup \{0\} \end{aligned}$$

- 1) **10 points.** Plot the constraints, feasible region, and gradient for this IP.
- 2) **25 points.** Solve the IP using the branch-and-bound method. In your solution, in addition to all calculations made, include a final version of the complete tree, where each node indicates the values of objective function and variables, and each branch shows the added constraints.
- 3) **25 points.** Solve the IP using the cutting plane method (with Gomory cuts).
- 4) **40 points.** Solve the linear relaxation (LP) of the IP using Dantzig-Wolfe decomposition (with Column generation). From the possible ways of decomposing the problem, select the one that would be more promising, following the criteria discussed in class (regarding the structure of the matrix of constraints). Then, write the reformulation, assuming no vertices from constraints A1 are known (so write a formulation using a generic index for the vertices). Then, write the associated master and auxiliary problems to be used in the column generation approach. To initialize the master problem, use either big M or two-phase method, assuming no columns had been created before. Finally, solve to optimality, clearly labeling and detailing each iteration.

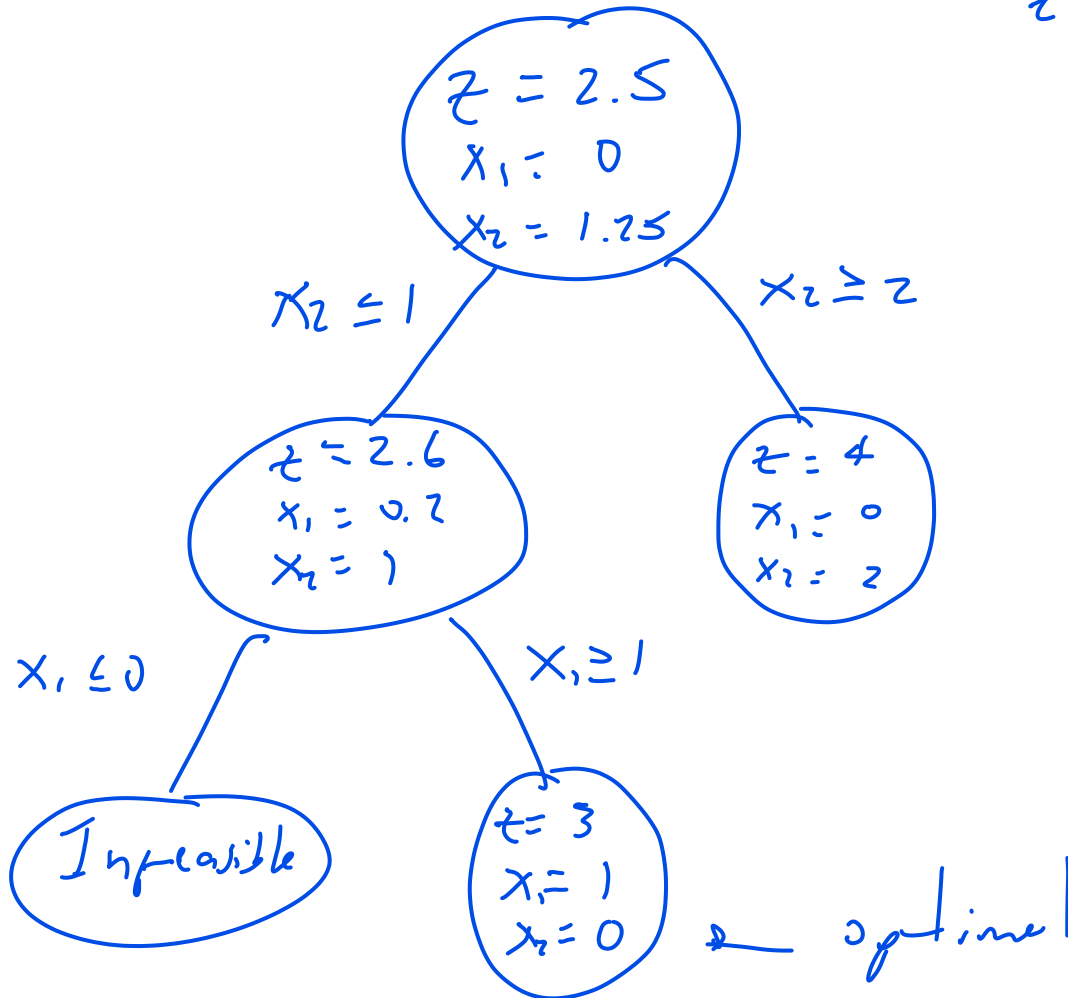
Solution:

- 1) **10 points.** Plot the constraints, feasible region, and gradient for this IP.



- 2) **25 points.** Solve the IP using the branch-and-bound method. In your solution, in addition to all calculations made, include a final version of the complete tree, where each node indicates the values of objective function and variables, and each branch shows the added constraints.

$$z = 3x_1 + 2x_2$$



3) 25 points. Solve the IP using the cutting plane method (with Gomory cuts)

$$B^{-1}A = \begin{pmatrix} 1.25 & 1 & -0.25 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ -1.25 & 0 & 0.25 & 0 & 1 \end{pmatrix}$$

$$B^{-1}b = \begin{pmatrix} 1.25 \\ 3 \\ 0.75 \end{pmatrix} \begin{matrix} x_2 \\ s_2 \\ s_3 \end{matrix}$$

cut 1

$$\Rightarrow 1.25x_1 - 0.25s_1 = 1.25$$

$$\Rightarrow 0.25x_1 + 0.75s_1 \geq 0.75$$

$$\Rightarrow \frac{1}{4}x_1 + \frac{3}{4}(-5 + 5x_1 + 4x_2) \geq \frac{1}{4}$$

$$\Rightarrow \frac{16}{4}x_1 + \frac{12}{4}x_2 \geq \frac{16}{4}$$

$$\Rightarrow 4x_1 + 3x_2 \geq 4$$

cut 2

$$\Rightarrow -1.25x_1 + 0.25s_1 + s_2 = 0.75$$

$$\Rightarrow 0.75x_1 + 0.25s_1 \geq 0.75$$

$$\Rightarrow \frac{3}{4}x_1 + \frac{1}{4}(-5 + 5x_1 + 4x_2) \geq \frac{3}{4}$$

$$\Rightarrow \frac{8}{4}x_1 + \frac{4}{4}x_2 \geq \frac{8}{4}$$

$$\Rightarrow 2x_1 + x_2 \geq 2$$

with these cuts, the solution to the new problem is $x_1 = 1, x_2 = 0$

since it's integer, it's optimal! $\Rightarrow z = 3$

4) **40 points.** Solve the linear relaxation (LP) of the IP using Dantzig-Wolfe decomposition (with Column generation). From the possible ways of decomposing the problem, select the one that would be more promising, following the criteria discussed in class (regarding the structure of the matrix of constraints). Then, write the reformulation, assuming no vertices from constraints A1 are known (so write a formulation using a generic index for the vertices). Then, write the associated master and auxiliary problems to be used in the column generation approach. To initialize the master problem, use either big M or two-phase method, assuming no columns had been created before. Finally, solve to optimality, clearly labeling and detailing each iteration.

$$\begin{aligned} \text{LP} = \min \quad & 3x_1 + 2x_2 \\ \text{s.t.} \quad & 5x_1 + 4x_2 \geq 5 \\ & x_1 \leq 3 \\ & x_2 \geq 2 \\ & x_1, x_2 \geq 0 \end{aligned}$$

$$A_0 = \begin{bmatrix} 5 & 4 \end{bmatrix}; b_0 = 5$$

$$A_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}; b_1 = 3$$

$$\Rightarrow \min (3 \ 2) \sum_{i \in L} \pi_i v_i$$

$$\text{s.t.} \quad (5 \ 4) \sum_{i \in L} \pi_i v_i \geq 5$$

$$\sum_{i \in L} \pi_i = 1$$

RMP

$$\min (3 \ 2) \sum_{i \in L_K} \pi_i v_i$$

$$(5 \ 4) \sum_{i \in L_K} \pi_i v_i \geq 5$$

$$\sum_{i \in L_K} \pi_i = 1$$

A.P.

$$\max (\pi^T A_0 - c^T) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$\text{s.t.} \quad x_1 \leq 3$$

$$x_2 \leq 2$$

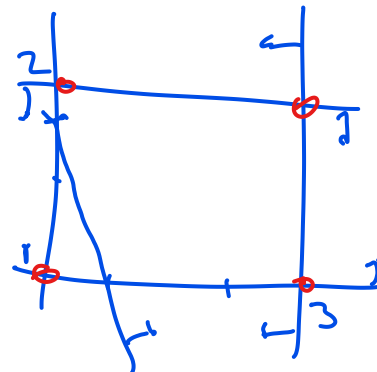
$$x_1, x_2 \geq 0$$

$$\Rightarrow \text{AP} \max \left[\pi^T (5 \ 4) - (3 \ 2) \right] \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$\text{s.t.} \quad x_1 \leq 3$$

$$x_2 \leq 2$$

$$x_1, x_2 \geq 0$$



Iteration 1

$$V = \{ \}$$

↳ We use big-M to initialize

MP

$$\min Mv_1 + Mv_2$$

$$\text{s.t. } v_1 - s_1 = 5$$

$$v_2 = 1$$

$$v_1, v_2, s_1 \geq 0$$

$$B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow C_B^T B^{-1} = (M \ M) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = (M \ M)$$

$$\Rightarrow \text{Aux} \quad \max \left[M \begin{pmatrix} 5 & 4 \end{pmatrix} - (3 \ 2) \right] \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = (5M-3)x_1 + (4M-2)x_2$$

$$\text{s.t. } x_1 \leq 3$$

$$x_2 \leq 2$$

$$x_1, x_2 \geq 0$$

$$\begin{aligned} &\Rightarrow x_1^* = 3; x_2^* = 2 \\ &\Rightarrow z^* = 15M - 7 + 8M - 4 = 23M - 11 \\ &\Rightarrow r_1 = (23M - 11) + M = 24M - 11 > 0 \\ &\Rightarrow \text{enters MP} \end{aligned}$$

Iteration 2

$$\Rightarrow P_{\lambda_1} = \left[\begin{pmatrix} 15 & 4 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \end{pmatrix} \right] = \begin{bmatrix} 23 \\ 1 \end{bmatrix} \Rightarrow \boxed{\text{MP}}$$

$$C_{\lambda_1} = (2 \ 2) \begin{pmatrix} 3 \\ 2 \end{pmatrix} = 13$$

$$\Rightarrow \boxed{\text{AP}} \quad \max \left[0 \begin{pmatrix} 5 & 4 \end{pmatrix} - (3 \ 2) \right] \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = -3x_1 - 2x_2$$

$$\text{s.t. } x_1 \leq 3$$

$$x_2 \leq 2$$

$$x_1, x_2 \geq 0$$

$$\Rightarrow x_1^* = x_2^* = 0$$

$$\Rightarrow z^* = 0$$

$$\Rightarrow r_{\lambda_1} = 0 + 13 = 13 > 0$$

$$\Rightarrow \text{enters MP}$$

$$\min Mv_1 + Mv_2 + 13\lambda_1$$

$$\text{s.t. } v_1 + 23\lambda_1 - s_1 = 5$$

$$v_2 + \lambda_1 = 1$$

$$v_1, v_2, \lambda_1, s_1 \geq 0$$

$$\Rightarrow \lambda_1^* = 1, v_1^* = 0, v_2^* = 0, s_1^* = 23 - 5 = 18$$

$$z^* = 13$$

$$\Rightarrow B = \begin{pmatrix} 23 & -1 \\ 1 & 0 \end{pmatrix}; C_B^T = (13 \ 0)$$

$$\Rightarrow B^{-1} = \frac{1}{1} \begin{pmatrix} 0 & 1 \\ -1 & 23 \end{pmatrix} \Rightarrow C_B^T B^{-1} = (0 \ 13)$$

Iteration 3

$$\Rightarrow P_{\lambda_2} = \left[\begin{pmatrix} 15 & 4 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right] = \begin{bmatrix} 0 \\ 0 \end{bmatrix}; C_{\lambda_2} = (3 \ 2) \begin{pmatrix} 0 \\ 0 \end{pmatrix} = 0 \Rightarrow \boxed{\text{MP}}$$

$$\Rightarrow \boxed{\text{AP}} \quad \max \left[\frac{13}{23} \begin{pmatrix} 5 & 4 \end{pmatrix} - (3 \ 2) \right] \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \frac{65-69}{23} x_1 + \frac{52-46}{23} x_2$$

$$\text{s.t. } x_1 \leq 3$$

$$x_2 \leq 2$$

$$x_1, x_2 \geq 0$$

$$\Rightarrow x_1^* = 0; x_2^* = 2; z^* = \frac{6+2}{23}$$

$$\Rightarrow r_{\lambda_2} = \frac{13}{23} + 0 > 0$$

$$\Rightarrow \text{enters MP}$$

$$\min Mv_1 + Mv_2 + 13\lambda_1 + 0\lambda_2$$

$$\text{s.t. } v_1 + 23\lambda_1 + 0\lambda_2 - s_1 = 5$$

$$v_2 + \lambda_1 + \lambda_2 = 1$$

$$v_1, v_2, \lambda_1, \lambda_2, s_1 \geq 0$$

$$\Rightarrow \lambda_1^* = 5/23; \lambda_2^* = 1 - \lambda_1^* = 18/23$$

$$v_1^* = v_2^* = s_1^* = 0 \Rightarrow z^* = 13 \times \frac{5}{23}$$

$$\Rightarrow B = \begin{pmatrix} 23 & 0 \\ 1 & 1 \end{pmatrix}; C_B^T = (0 \ 0)$$

$$B^{-1} = \frac{1}{23} \begin{pmatrix} 1 & -1 \\ 0 & 23 \end{pmatrix} \Rightarrow C_B^T B^{-1} = \left(\frac{13}{23} \ 0 \right)$$

Iteration 4

$$\Rightarrow P_{\lambda_3} = \left[\begin{pmatrix} 15 & 4 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right] = \begin{bmatrix} 0 \\ 0 \end{bmatrix}; C_{\lambda_3} = (3 \ 2) \begin{pmatrix} 0 \\ 0 \end{pmatrix} = 0 \Rightarrow \boxed{\text{MP}}$$

$$\Rightarrow \boxed{\text{AP}} \quad \max \left[\frac{1}{2} \begin{pmatrix} 5 & 4 \end{pmatrix} - (3 \ 2) \right] \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \frac{5-6}{2} x_1 + \frac{4-4}{2} x_2$$

$$\text{s.t. } x_1 \leq 3$$

$$x_2 \leq 2$$

$$x_1, x_2 \geq 0$$

$$\Rightarrow x_1^* = 0; x_2^* \in [0, 1]; z^* = 0 \Rightarrow r_{\lambda_3} = 0 + 0 = 0 \Rightarrow \lambda_3 \text{ not enter MP}$$

$$\min Mv_1 + Mv_2 + 13\lambda_1 + 0\lambda_2 + 0\lambda_3$$

$$\text{s.t. } v_1 + 23\lambda_1 + 0\lambda_2 + 0\lambda_3 - s_1 = 5$$

$$v_2 + \lambda_1 + \lambda_2 + \lambda_3 = 1$$

$$v_1, v_2, \lambda_1, \lambda_2, \lambda_3, s_1 \geq 0$$

$$\Rightarrow \lambda_3^* = \frac{5}{8}; \lambda_2^* = 1 - \lambda_3^* = \frac{3}{8}; v_1^* = v_2^* = s_1^* = 0 \Rightarrow z^* = 9 \times \frac{5}{8}$$

$$\Rightarrow B = \begin{pmatrix} 0 & 8 \\ 1 & 1 \end{pmatrix}; C_B^T = (0 \ 4) \Rightarrow B^{-1} = \frac{1}{8} \begin{pmatrix} 1 & -8 \\ 0 & 8 \end{pmatrix} \Rightarrow C_B^T B^{-1} = \left(\frac{1}{2} \ 0 \right)$$

$$\left[\frac{5}{2} \right]$$

$$\Rightarrow \text{current solution is optimal} \Rightarrow z^* = \frac{5}{2}; \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \frac{5}{8} \lambda_3; v_1 = \frac{3}{8} \begin{pmatrix} 0 \\ 0 \end{pmatrix} + \frac{5}{8} \begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 5/4 \end{pmatrix} = \begin{pmatrix} 0 \\ 5/4 \end{pmatrix}$$